Notes on Gravitation and Relativity

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Chapter 1

Some Mathematical Preliminaries

1.1 Introduction

This chapter introduces some fundamental mathematics that will be needed in the later chapters. The terms appearing in this chapter should be familiar to any student of physics. However, due to the intuitive way these terms are usually presented in any first course of calculus at the university freshman level, students are not always familiar with the more "mathematical" way of presenting these concepts. Since in the later study of basic differential geometry these will work as a necessary prerequisite vocabulary, it is better to be familiar with the language a bit.

Chapter 2

Manifolds

Chapter 3

Derivatives on Manifolds

3.1 Introduction

In this chapter, we use some of the concepts introduced in the previous one to talk about the very fascinating concept of differentiation on a manifold. We all have an idea about what a differentiation is. It is a small change in something over small change in some parameter value. For example, the x component of velocity is the change in position on the x axis, Δx over change in the parameter time, Δt. When taking a derivative of a tensor field over a manifold, we need to carefully define what change in the tensor is and what parameter comes in the denominator. It turns out that there are at least three interesting ways we can define a change in a tensor, and they will give rise to three different types of derivatives.

Now the naive way of taking the change of a tensor T between nearby points P and Q is to simply take the difference, T(Q)-T(P). This does not work for the simple reason that P and Q are different points, even if they are close, and as such, the tensors T(Q) and T(P) simply live in completely different vector spaces. It just is not possible to take their difference.

This gives rise to the need to introduce a connecting mechanism that will connect the tensor at T(Q) and give a unique counterpart of T(Q) at the point P, call it T′(P), which belongs to the same vector space as T(P). Now we can define the difference as T′(P)-T(P).

3.2 Lie Derivative

Here, the connecting mechanism comes from a smooth vector field, X. The local 1 parameter group of diffeomorphism generated by X introduce flow lines. The two points p and q will lie close together and on the same flow line.

This is how the definition goes,

Definition 3.1 (Lie Derivative)**.** Let T be a tensor field over M and p be a point in M. Let X be a vector field and $\{\Gamma_t : U \longrightarrow \Gamma_t[U]\}_{t\in I}$ be a local 1 parameter family of diffeomorphism generated by X, where U is an open subset of M and p$\in$ U, and I is an open subset of $\mathbb{R}$. Then the Lie derivative of T with respect to X, at the point p is defined as,
$$\pounds_X T = \lim_{t\to 0} \frac{(\Gamma_t)^* T - T}{t}$$

I have avoided writing $\Big|_p$ on all of the tensors to avoid clutter, and I will consistently do this kind of abuse of notation but it will be clear from context.

The right hand side of the definition tells us how the mechanism of connecting back. To evaluate the Lie derivative of T wrt X at the point p, I first go parameter distance t away from the point p along the maximal integral curve of X with initial point p, that is, at the point $\Gamma_t(p)$. There, the tensor T$\Big|_{\Gamma_t(p)}$ is taken. Then using the pull-back map $(\Gamma_t)^*$ I pull that tensor back to the point p, making $(\Gamma_t)^* T\Big|_p$. Then I subtract from this the tensor T$\Big|_p$. This is how the difference is defined. And the parameter in the denominator is simply the parameter distance t along the maximal integral curve of X with initial point p.

Proposition 3.2. The operator $\pounds_X$ has the following properties.

1. Commutes with addition. $\pounds_X(T + S) = \pounds_X T + \pounds_X S$
2. Satisfies the Leibniz rule. $\pounds_X(T \otimes S) = (\pounds_X T) \otimes S + T \otimes (\pounds_X S)$
3. Commutes with index substitution and contraction

Chapter 4

Submanifolds and Hypersurfaces

4.1 Introduction

This is a crucial ingredient in understanding almost anything that has to do with Numerical Relativity, the art of solving EFEs by the help of numerical methods. I will closely follow the treatment given by David Malament in his book on the foundations of GR and Newtonian gravity. In my opinion, among all the GR textbooks, this one has the most systematic treatment of this topic. Though, being written by a philosopher, the main idea of the book is not to teach differential geometry so that one can use it in research, rather it is to present only that much which forms the core of our understanding of the topic. However, this much is actually sufficient for diving deep into a proper numerical relativity textbook.

4.2 Imbedding

Let S and M be two manifolds. Let there be a smooth map $\Psi : S \longrightarrow M$. Ψ is called an imbedding if the following three conditions hold.

- Ψ is injective.
- at all p in S, the push forward map $(\Psi_p)_* : T_pS \longrightarrow T_pM$ is injective.
- the inverse map $\Psi^{-1} : \Psi[S] \longrightarrow S$ is continuous with respect to the relative topology in $\Psi[S]$.

At first glance, this definition might seem unmotivated and built out of thin air. TODO: learn and write the reasons behind this definition

Now, suppose that S⊆M, then we say that S is an imbedded submanifold of M if the map $id : S \longrightarrow M$ is an imbedding. If the dimension of M is n, and the dimension of S is k, and $k = n - 1$ then we say that S is a *hypersurface* in M.

In this and the next few sections, let me fix S to be a k-dimensional imbedded submanifold of M. Later on we will think more specifically in terms of hypersurfaces. But for now let me go through some mathematical machinary needed for imbedded submanifolds in general.

4.3 S-tensors vs M-tensors

At a point p, we can talk about tensors with respect to the atlas of M, or with respect to the atlas of S. We refer to the former as M-tensors, and the later as S-tensors. The S-tensors are S's own objects and we do not need to refer to M at all while talking about them. We will use a tilde over the name of the tensor to show that it is an S-tensor.

Given S to be a metric submanifold of M (I will explain what a metric submanifold is in a bit), we can divide T_pM into two subspaces, one would be a subspace of vectors "tangent" to S, and one "normal" to S.

So what, then, is an M-vector at p that is "tangent" to S? (I am using quotes because this tangent is not related to the word "tangent" in "tangent vector").

Definition 4.1. A vector $X \in T_pM$ is *tangent* to S if $X \in (id_p)_*(T_pS)$.

Definition 4.2. A covector $\eta \in T_p^*M$ is normal to S if for all vectors $X \in T_pM$ that are tangent to S, $\eta(X) = 0$

We denote the space of all M-vectors tangent to S at p as $\perp_p$ and the space of all M-covectors normal to S at p as N_p^*. As we took T_pS and then pushed it forward with an imbedding (which basically one to one maps T_pS to T_pM, the dimension of $\perp_p$ is k. We can also show that the dimension of N_p^* is $n-k$.

Another way to describe tangent vector comes from what it means to push forward a vector from one manifold to another. For a vector $\tilde{X}$ in S, there is obviously a curve γ to which the vector $\tilde{X}$ is tangent to. Since push forward of a vector tangent to a curve is tangent to the image of the curve, $id_*\tilde{X}$ is tangent to $id(\gamma)$. In fact, we are considering S to be a subset of M, and $id(\gamma)$ is actually just γ. This means, any M-vector at p that is tangent to a curve γ which lies entirely in S, is a memeber of $\perp_p$. Wait, how do you show the converse.....?????

Proposition 4.3. The dimension of N_p^* is $n-k$.

Proof. First of all, we know that dimension of $\perp_p$ is k. Take k vectors $\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}$ in $\perp_p$ that are linearly independent. Now, to make a basis of T_pM we need some other linearly independent vectors outside of $\perp_p$. Let's say the total basis is $\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}, \overset{k+1}{X}, ..., \overset{n}{X}$. Now, let us take the dual basis of the previous basis :

$\overset{1}{\eta}, \overset{2}{\eta}, ..., \overset{k}{\eta}, \overset{k+1}{\eta}, ..., \overset{n}{\eta}$. Then, the covectors $\overset{k+1}{\eta}, ..., \overset{n}{\eta}$ span a space N' with the following property: any member η of N', being a combination of $\overset{k+1}{\eta}, ..., \overset{n}{\eta}$, will actually produce $\eta(X) = 0$ for any member X of $\perp_p$ as X is a combination of $\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}$. That means the space N'. Clearly, N' is actually what we are calling N_p^*. And the dimension is $n - l$ □

Once we have N_p^* defined, we can prove the following proposition, which will be very useful later on where we use this result at the same rate, or even more, than the first definition of a vector being tangent.

Proposition 4.4. An M-tensor X is tangent to S at p iff for any $\eta \in N_p^*$, $\eta(X) = 0$.

Proof. Let us use the basis vectors as defined before, that is, T_pM is spanned by $\{\overset{1}{X}, \overset{2}{X}, ..., \overset{k}{X}, \overset{k+1}{X}, ..., \overset{n}{X}\}$ where the first k spans $\perp_p$. Let T_p^*M be spanned by the dual vectors $\{\overset{1}{\eta}, \overset{2}{\eta}, ..., \overset{k}{\eta}, \overset{k+1}{\eta}, ..., \overset{n}{\eta}\}$ where the last $n - k$ vectors span N_p^*.

Now, let $X \in \perp_p$.
This means X has the form

$$X = \sum_1^k a_j \overset{j}{X}$$

For any $\eta \in N_p^*$, η has the form,

$$\eta = \sum_{k+1}^n b_j \overset{j}{\eta}$$

Clearly, $\eta(X) = 0$.

Lets now assume that we have an $X \in T_pM$ such that for any $\eta \in N_p^*$, $\eta(X) = 0$. This means in the expansion of X, there cannot be any term of the form $a_j\overset{j}{X}$ where $j > k$. And so X is a linear combination of only the basis vectors $\{\overset{1}{X}, ...\overset{k}{X}\}$. And that means, $X \in \perp_p$

□

While the M-vectors tangent to S come directly from S via a push forward map, the M-covectors pulled back to S actually give zero.

Proposition 4.5. $\eta \in N_p^* \iff (id_p)^*(\eta) = 0$

Proof. Let $\eta \in N_p^*$
Let $\tilde{X} \in T_pS$. So, $((id_p)^*(\eta))\left(\tilde{X}\right) = \eta\left((id_p)_*\tilde{X}\right) = 0$. Since this is true for any $\tilde{X}$ in T_pS, we can conclude that $(id_p)^*(\eta) = 0$.

Now, let $(id_p)^*(\eta) = 0$ where $\eta \in T_p^*M$
For any $\tilde{X} \in T_pS$,

$$((id_p)^*(\eta))\,(\tilde{X}) = 0$$

$$\Longrightarrow \eta\left((id_p)_*(\tilde{X})\right) = 0$$

this means η acting on any member of $\perp_p$ is zero. And that means $\eta \in N_p^*$.

□

Now, more general M-tensors at p can also be tangent and normal, in different indices. For example, we can say when a (2,3) M-tensor T^{ab}_{cde} at p is tangent in the first upper index a, or normal in the third lower index e. The definition that will follow can be easily generalized to the most general case, but I will only give a restricted special case definition, because writing down the most general one is more annoying than illuminating.

Definition 4.6. A (2,3) M-tensor $T^{ab}{}_{cde}$ at point p is tangent to S in the second upper index b if for any $\eta \in N_p^*$, $T^{ab}{}_{cde}\eta_b = 0$. It is normal to S in the third lower index e if for any $X \in \perp_p$, $T^{ab}{}_{cdn}X^n = 0$. And so on for tensors of different index structures.

So far we are only considering being tangent on an upper index, and being normal on a lower index. But it is also possible to be normal on an upper index and to be tangent on a lower index if the manifold M is a metric manifold (even at this point I do not need the submanifold to have a metric).

Definition 4.7. $T^{ab}{}_{cde}$ is normal in the first upper index a if $T_a{}^b{}_{cde} = g_{am}T^{mb}{}_{cde}$ is normal in the index a. Similarly T is tangent in the second lower index d if $T^{ab}{}_c{}^d{}_e = g^{dm}T^{ab}{}_{cme}$ is tangent in the index d. And so on for tensors of other index structure.

Basically, this definition says, we need to raise or lower an index. Then if after raising a lower index the tensor is tangent, then it is tangent in that index in it's lower form too. And if after lowering an index, the tensor is normal, then that tensor is normal in that index in the raised form too.

Right now, for M-vectors and M-covectors at p, we have 4 different spaces in terms of tangency and normalcy to S.

- $\perp_p$ is the space of M-vectors tangent to S at p.
- $\perp_p^*$ is the space of M-covectors tangent to S at p.
- N_p is the space of M-vectors normal to S at p.
- N_p^* is the space of M-covectors normal to S at p.

We know that the dimension of $\perp_p$ is k and the dimension of N_p^* is $(n-k)$. What is the dimension of $\perp_p^*$? Or N_p ?

Proposition 4.8. The dimesion of $\perp_p^*$ is k and the dimension of N_p is $(n-k)$.

Proof. We know that the metric tensor g_{ab} creates a bijective mapping between T_pM and T_p^*M. So thinking from that perspective, $\perp_p^*$ is basically the set of all covectors X such that $X_a = g_{am}X^m$ where $X \in \perp_p$. And the dimension of such a space is obviously the dimension of $\perp_p$, that is, k.

In a similar way, the space N_p is made of all vectors η that has the form $\eta^a = g^{am}\eta_m$ where $\eta \in N_p^*$. And the dimension of such a space is equal to the dimension of N_p^*, which is $(n-k)$. □

Now we might be thinking of a nice possibility. Maybe, $T_pM = \perp_p + N_p$

4.4 Sticking to Only M-Tensors

At any point p, there is a natural isomorphism between S-tensors of any index structure and M-tensors of the same index structure that are tangent to S. This is the very basis of how we talk about a hypersurface. The idea is that we will commit to M-tenosrs tangent to S and not talk about S-tensors.

Let me first describe the isomorphism. Here I will satisfy myself by only defining the map, and I will skip the proof of the map being an isomorphism.

The map ϕ_p takes an S-tensor $\tilde{X}^{ab}{}_{cd}$ to $\phi_p\left(\tilde{X}^{ab}{}_{cd}\right)$, which is an M-tensor. We define the isomorphism in stages, first for scalars, then for vectors, covectors and finally for tensors of arbitray index structure.

First, for a scalar α at p, $\phi_p(\alpha) = \alpha$. There is no need to use tilde as scalars are just numbers.

Second, for an S-vector $\tilde{X}^a$ at p, $\phi_p\left(\tilde{X}^a\right) = (id_p)_* \tilde{X}^a$. By the very definition of a vector being tangent to S at p, $\phi_p\left(\tilde{X}^a\right)$ is tangent to S.

Third, for a S-covector $\tilde{\eta}_a$ at p, $\phi_p(\tilde{\eta}_a)$ is defined by its action on any M-vector X^a.

$$\phi_p(\tilde{\eta}_a) X^a = \begin{cases} \tilde{\eta}_a\left(\phi_p^{-1}X^a\right); & X \in \perp_p \\ 0; & X \in N_p \end{cases}$$

We remark that this is well defined, as $\phi_p^{-1}X^a$ is well defined. Also, notice that the second clause of the definition of $\phi_p(\tilde{\eta}_a)$ ensures that it is tangent to S.

Lastly, lets take an arbitray index structure S-tensor $\tilde{X}^{ab}{}_c$, then,

$$\phi_p\left(\tilde{X}^{ab}{}_c\right)A_aB_bC^c = \begin{cases} \tilde{X}^{ab}{}_c\left(\phi_p^{-1}A_a\right)\left(\phi_p^{-1}B_b\right)\left(\phi_p^{-1}C^c\right); & A_a, B_b, C^c \text{ are tangent to S} \\ 0; & \text{if } A_a, B_b, \text{or } C^c \text{ is normal to S} \end{cases}$$

Again, $\phi_p\left(\tilde{X}^{ab}{}_c\right)$ is tangent to S by the second clause above.

One can also show that ϕ_p commutes with all tensor operations, i.e. addition, outer multiplication, index substitution, contraction.

For example,

$$\phi_p\left(\tilde{X}^a\tilde{Y}^b\right) = \phi_p\left(\tilde{X}^a\right)\phi_p\left(\tilde{Y}^b\right)$$

In short, at any point p in S, the tensor algebra of S-tensors is isomorphic to the tensor algebra of M-tensors that are tangent to S. With every S-tensor, I get an evil twin M-tensor of the same index structure.

Now, this can be extended to the whole of S, and we have a map ϕ that takes any S-tensor field to an M-tensor field that is tangent to S and have the same index structure, and they also follow the same tensor algebra on both sides of the "river".

S-smoothness and M-smoothness

First of all, let us recall a very handy notion of the smoothness of a tensor field. Say, we have a manifold M, and an atlas $\mathcal{A}$. Now, a tensor field $T^a_{\ b}$ is smooth iff the components are smooth functions with respect to any coordinatization of the manifold with charts from the atlas $\mathcal{A}$. This means the idea of smoothess is tied to the atlas of a manifold.

Now think of the manifold S. Let me define by O an open set of M that covers S, i.e. S is a subset of O. This is going to be our working definition of O for quite some time in this discussion, unless otherwise specified. An M-tensor field T tangent to S can be smooth in two senses. If an extension of T on O, named $\overset{+}{T}$, is smooth wrt $\mathcal{A}_M$, then T is M-smooth. If, on the other hand, the S-tensor field $\phi^{-1}T$ is smooth wrt $\mathcal{A}_S$, then T is S-smooth. Now, there is a theorem that says for any M-tensor field T tangent to S, it is M-smooth iff it is S-smooth. So saying that T (when it is tangent to S) is smooth means it is smooth in both ways. On the other hand if T is not tangent to S, then it can only be M-smooth.

h_{ab}

This is the star of the show. We call it a projector for reasons to become clear soon. Let us define it first. Say there is a metric g_{ab} on M. We define the "induced metric" or "first fundamental form" on S by taking the pull back of g_{ab} wrt to the identity map. The induced metric, an S-field, $\tilde{h}_{ab} = id^* g_{ab}$. Then, we define the smooth symmetric M-tensor h_{ab} which is tangent to S as the ϕ-image of this induced metric $\tilde{h}_{ab}$, so that $h_{ab} = \phi\left(\tilde{h}_{ab}\right) = \phi\left(id^* g_{ab}\right)$.

This is as counterintuitive and hard to remember as possible. So we actually use an equivalent definition of h_{ab}. Here, we define h_{ab} by the way it acts on M-vectors.

$$h_{ab}X^aY^b = \begin{cases} g_{ab}X^aY^b; & \text{if } X, Y \text{ are tangent to S} \\ 0; & \text{if any of } X, Y \text{ are normal to S} \end{cases} \tag{4.1}$$

Let me show that the equivalence works. Lets say both X and Y are tangent to S. Then,

$$\begin{aligned} h_{ab}X^aY^b &= \phi\left(id^* g_{ab}\right) X^aY^b \\ &= \left(id^* g_{ab}\right)\left(\phi^{-1}X^a\right)\left(\phi^{-1}Y^b\right) \\ &= g_{ab}\; id_*\left(\phi^{-1}X^a\right)\; id_*\left(\phi^{-1}Y^b\right) \\ &= g_{ab}\; \phi\left(\phi^{-1}X^a\right)\phi\left(\phi^{-1}Y^b\right) \\ &= g_{ab}X^aY^b \end{aligned}$$

If any of X or Y is not tangent, then by virtue of h_{ab} being tangent to S, $h_{ab}X^aY^b$ is 0.

We define a perpendicular projection tensor k_{ab} too which is a smooth symmetric M-tensor field.

$$k_{ab} = g_{ab} - h_{ab} \tag{4.2}$$

Notice that k_{ab} is normal to S. If X is a M-vector tangent to S, then, $k_{ab}X^a = g_{ab}X^a - h_{ab}X^a = g_{ab}X^a - g_{ab}X^a = 0$. So k_{ab} is normal to S in the first index. But being a symmetric tensor, it must be also normal in the second index too, so it is normal as a whole.

The following propositions show that it is okay to call h_{ab} a tangential projector and k_{ab} a perpendicular or normal projector. notice that k_{ab} is normal to S.

Proposition 4.9. X^a is tangent to S $\iff h^a{}_b X^b = X^a \iff k^a{}_b X^b = 0$

Proof. We will show the first equivalence and then the second equivalence.

First equivalence:

Let X be tangent to S.

I will show that the action of X^a and $h^a{}_b X^b$ is the same on any M-covector η_a. If η_a is normal to S, then both sides give zero. If η_a is tangent to S, then $h^a{}_b X^b \eta_a = h_{ab} \eta^a X^b = g_{ab} \eta^a X^b = X^a \eta_a$.

Conversely, let $h^a{}_b X^b = X^a$. I need to show that X^a is tangent to S. Take η_a normal to S. Then $\eta_a X^a = \eta_a h^a{}_b X^b = 0$. The last equality follows from the fact that h_{ab} is tangent in both indices.

This completes the proof for the first equivalence.

Second equivalence:

Let $h^a{}_b X^b = X^a$.

Then, $k^a{}_b X^b = \left(g^a{}_b - h^a{}_b\right) X^b = X^a - X^a = 0$

Conversely, let, $k^a{}_b X^b = 0$.

Then, $h^a{}_b X^b = \left(g^a{}_b - k^a{}_b\right) X^b = X^a - 0 = X^a$

This completes the proof of the second equivalence.

□

Proposition 4.10. X^a is normal to S $\iff k^a{}_b X^b = X^a \iff h^a{}_b X^b = 0$

Proof. Similar as before. □

While I stated proposition 4.9 and 4.10 for a (1,0) tensor field, i.e. a vector field X^a, these propositions actually hold for tensor fields of arbitrary index structure. For example, if $T^{ab}{}_c$ is tangent to S in the second upper index, then $h^b{}_m T^{am}{}_c = T^{ab}{}_c$ and $k^b{}_m T^{am}{}_c = 0$.

Proposition 4.11. $h^a{}_b h^b{}_c = h^a{}_c$

Let me pause here for a moment, and fill up the rest of the proofs up to the Gauss Codazzi equations

Extrinsic Curvature

We know that,

$$h^m{}_a h^n{}_b h^p{}_c \nabla_m h_{np} = 0$$

$$h^m{}_a k^n{}_b k^p{}_c \nabla_m h_{np} = 0$$

However, the hhk projection, $h^m{}_a h^n{}_b k^p{}_c \nabla_m h_{np}$ is not zero and we define a new quantity, called the extrinsic curvature of S, as follows,

$$K_{abc} = h^m{}_a h^n{}_b k^p{}_c \nabla_m h_{np}$$

K_{abc} has particular geometric significance, and we will show some of those significance in this section.

Gauss Codazzi Equations

Before proving those equations, let me remind one useful way of characterizing the Riemann tensor. If v^a is a vector field, then,

$$\begin{aligned}
[\nabla_c \, , \nabla_d] v^a &= R^a{}_{mcd} v^m \\
2\nabla_{[c} \nabla_{d]} v^a &= R^a{}_{mcd} v^m \\
\nabla_{[c} \nabla_{d]} v^a &= \frac{1}{2} R^a{}_{mcd} v^m
\end{aligned}$$

This can also be generalized, in the following way. Say, T^a_b is a (1,1) field. Then the antisymmetrized double covariant derivative on it can be expanded in temrs of the Riemann tensor as follows,

$$2\nabla_{[c} \nabla_{d]} T^a_b = R^a{}_{mcd} T^m_b - R^m{}_{bcd} T^a_m$$

One final fact about the Riemann tensors that needs a reminder is, $R_{abcd} = R_{cdab}$.

We will use a script R, $\mathcal{R}$, to denote the Riemann tensor defined wrt to D. This is going to be a tangent tensor field. So, we define the Riemann tensor to be, $\mathcal{R}^a{}_{mcd} v^m = 2D_{[c} D_{d]} v^a$ where v^a is an M-field tangent to S.

To Do: Write down some more words about the proper definition of $\mathcal{R}$

Now,

$$
\begin{aligned}
D_{[c}D_{d]}v^a &= h^m{}_{[c}h^n{}_{d]}h^a{}_p\nabla_m D_n v^p \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_p\nabla_m\left(h^q{}_n h^p{}_r\nabla_q v^r\right) \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_p\left(\nabla_m h^q{}_n\right)h^p{}_r\nabla_q v^r + h^m{}_{[c}h^n{}_{d]}h^a{}_p h^q{}_n\left(\nabla_m h^p{}_r\right)\nabla_q v^r \\
&+ h^m{}_{[c}h^n{}_{d]}h^a{}_p h^q{}_n h^p{}_r\nabla_m\nabla_q v^r \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h^q{}_n\right)\nabla_q v^r + h^m{}_{[c}h^q{}_{d]}h^a{}_p\left(\nabla_m h^p{}_r\right)\nabla_q v^r \\
&+ h^m{}_{[c}h^q{}_{d]}h^a{}_r\nabla_m\nabla_q v^r \\
&\left(\text{Mostly using } h^a{}_b h^b{}_c = h^a{}_c\right)
\end{aligned}
\tag{4.3}
$$

Now, let's look at them term by term.

First Term:

$$
\begin{aligned}
& h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h^q{}_n\right)\nabla_q v^r \\
&= h^m{}_{[c}h^n{}_{d]}h^a{}_r\left(\nabla_m h_{nq}\right)\nabla^q v^r \\
&= h^a{}_r h^m{}_{[c}h^n{}_{d]}\left(\nabla_m h_{nq}\right)\nabla^q v^r \\
&= -h^a{}_r K_{[cd]q}\nabla^q v^r \\
&(\text{Using the relation } K_{abc} = -h^m{}_a h^n{}_b\nabla_m h_{nc}) \\
&= 0 \\
&(\text{ Since the antisymmetrization of the first two indices of } K_{abc} \text{ gives zero.})
\end{aligned}
$$

Second Term:

$$
\begin{aligned}
& h^m{}_{[c}h^q{}_{d]}h^a{}_p\left(\nabla_m h^p{}_r\right)\nabla_q v^r \\
&= h^m{}_{[c}h^q{}_{d]}h^{ap}\left(\nabla_m h_{pr}\right)\nabla_q v^r \\
&= h^m{}_{[c}h^q{}_{d]}h^{pa}\left(\nabla_m h_{pr}\right)\nabla_q v^r \\
&= g^{sa}h^m{}_{[c}h^q{}_{d]}h^p{}_s\left(\nabla_m h_{pr}\right)\nabla_q v^r \\
&= g^{sa}h^m{}_{[c}h^p{}_{|s|}h^q{}_{d]}\left(\nabla_m h_{pr}\right)\nabla_q v^r \\
&= -g^{sa}K_{[c|sr|}h^q{}_{d]}\nabla_q v^r \\
&= -g^{sa}K_{[c|sr|}h^q{}_{d]}\nabla_q\left(h^r{}_t v^t\right) \\
&(\text{The trick is to realize that v is tangent to S.}) \\
&= -K_{[c}{}^a{}_{|r|}h^q{}_{d]}\nabla_q\left(h^r{}_t v^t\right)
\end{aligned}
$$

$$
\begin{aligned}
&= -K_{[c}{}^{a}{}_{|r|} h^{q}{}_{d]} \left(\nabla_q h^{r}{}_{t}\right) v^{t} - K_{[c}{}^{a}{}_{|r|} h^{q}{}_{d]} h^{r}{}_{t} \nabla_q v^{t} \\
&= -K_{[c}{}^{a}{}_{|r|} h^{q}{}_{d]} \left(\nabla_q h^{r}{}_{t}\right) v^{t} - 0 \\
&\left(\begin{array}{l}\text{The second term goes to zero as the extrinsic curvature } K_{abc} \text{ is} \\ \text{normal in the third index.}\end{array}\right) \\
&= -K_{[c}{}^{a}{}_{|r|} h^{q}{}_{d]} \left(\nabla_q h^{r}{}_{t}\right) \left(h^{t}{}_{u} v^{u}\right) \\
&= -K_{[c}{}^{ar} h^{q}{}_{d]} \left(\nabla_q h_{tr}\right) \left(h^{t}{}_{u} v^{u}\right) \\
&= -K_{[c}{}^{ar} h^{q}{}_{d]} h^{t}{}_{u} \left(\nabla_q h_{tr}\right) v^{u} \\
&= K_{[c}{}^{ar} K_{d]ur} v^{u}
\end{aligned}
$$

Third Term:

$$
\begin{aligned}
& h^{m}{}_{[c} h^{q}{}_{d]} h^{a}{}_{r} \nabla_m \nabla_q v^{r} \\
&= h^{m}{}_{c} h^{q}{}_{d} h^{a}{}_{r} \nabla_{[m} \nabla_{q]} v^{r} \\
&= h^{m}{}_{c} h^{q}{}_{d} h^{a}{}_{r} \left(\frac{1}{2}\right) R^{r}{}_{nmq} v^{n} \\
&= \frac{1}{2} R^{r}{}_{nmq} h^{m}{}_{c} h^{q}{}_{d} h^{a}{}_{r} v^{n}
\end{aligned}
$$

Collecting the terms, I get from **??**,

$$
\begin{aligned}
D_{[c} D_{d]} v^{a} &= \frac{1}{2} R^{r}{}_{nmq} h^{m}{}_{c} h^{q}{}_{d} h^{a}{}_{r} v^{n} + K_{[c}{}^{ar} K_{d]ur} v^{u} \\
&= \frac{1}{2} R^{r}{}_{mnq} h^{n}{}_{c} h^{q}{}_{d} h^{a}{}_{r} v^{m} + K_{[c}{}^{ar} K_{d]mr} v^{m}
\end{aligned}
$$

Now, from our earlier characterization of $\mathcal{R}$, we have,

$$
\begin{aligned}
&\mathcal{R}^{a}{}_{mcd} v^{m} = 2 D_{[c} D_{d]} v^{a} \\
\implies &\mathcal{R}^{a}{}_{mcd} v^{m} = R^{r}{}_{mnq} h^{n}{}_{c} h^{q}{}_{d} h^{a}{}_{r} v^{m} + 2 K_{[c}{}^{ar} K_{d]mr} v^{m}
\end{aligned}
$$

Now, taking any vector η, not necessarily tangent, and then taking the tangential projection of that vector, $v^{m} = h^{m}{}_{s} \eta^{s}$,

$\mathcal{R}^a{}_{mcd}h^m{}_s\eta^s = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s \eta^s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s\eta^s$

$\mathcal{R}^a{}_{mcd}h^m{}_s\eta^s = \left(R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s\right)\eta^s$

$\mathcal{R}^a{}_{mcd}h^m{}_s = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]mr}h^m{}_s$

$\mathcal{R}^a{}_{scd} = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_s + 2K_{[c}{}^{ar}K_{d]sr}$

(We assumed that $\mathcal{R}$ is a tangent field, also K is tangent in the second index.)

$\mathcal{R}^a{}_{bcd} = R^r{}_{mnq}h^n{}_c h^q{}_d h^a{}_r h^m{}_b + 2K_{[c}{}^{ar}K_{d]br}$

$(s \to b)$

$\mathcal{R}^a{}_{bcd} = R^m{}_{rnq}h^n{}_c h^q{}_d h^a{}_m h^r{}_b + 2K_{[c}{}^{am}K_{d]bm}$

$(r \leftrightarrow m)$

$\mathcal{R}^a{}_{bcd} = R^m{}_{npq}h^p{}_c h^q{}_d h^a{}_m h^n{}_b + 2K_{[c}{}^{am}K_{d]bm}$

$(r \to n, n \to p)$

$\mathcal{R}^a{}_{bcd} = R^m{}_{npq}h^a{}_m h^n{}_b h^p{}_c h^q{}_d + 2K_{[c}{}^{am}K_{d]bm}$

And let me put this in a box, as I have to memorize this as the Make the box. Then start deriving the second Gauss Codazzi relation.

The second Gauss Codazzi equation is $h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m K_{abc} = \frac{1}{2}h^m{}_n h^a{}_p h^b{}_q h^c{}_r R_{mabc}$

To prove it, let's start with the LHS,

$$\begin{aligned}
LHS &= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m K_{abc} \\
&= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m \left(-h^s{}_a h^t{}_b \nabla_s h_{tc}\right) \\
&= -h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h^s{}_a\right) h^t{}_b \nabla_s h_{tc} - h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&- h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r h^s{}_a h^t{}_b \nabla_m \nabla_s h_{tc} \\
&\equiv -A - B - C \text{ (Naming the three terms.)}
\end{aligned}$$

Term A:

$$\begin{aligned}
A &= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h^s{}_a\right) h^t{}_b \nabla_s h_{tc} \\
&= h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \left(\nabla_m h_{as}\right) h^t{}_b \nabla^s h_{tc} \\
&= h^m{}_{[n}h^a{}_{p]}h^t{}_q k^c{}_r \left(\nabla_m h_{as}\right) \left(\nabla^s h_{tc}\right) \\
&= -K_{[np]s}h^t{}_q k^c{}_r \nabla^s h_{tc}
\end{aligned}$$

$$= 0$$

Term B:

$$\begin{aligned}
B &= h^m{}_{[n} h^a{}_{p]} h^b{}_q k^c{}_r h^s{}_a \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&= h^m{}_{[n} h^s{}_{p]} h^b{}_q k^c{}_r \left(\nabla_m h^t{}_b\right) \nabla_s h_{tc} \\
&= h^m{}_{[n} h^s{}_{p]} h^b{}_q k^c{}_r \left(\nabla_m h_{bt}\right) \nabla_s h^t{}_c \\
&= -K_{[n|qt|} h^s{}_{p]} k^c{}_r \nabla_s h^t{}_c \\
&= -k^u{}_t K_{[n|qu|} h^s{}_{p]} k^c{}_r \nabla_s h^t{}_c \\
&\text{(K is normal in the third index, so introducing k makes sense.)} \\
&= -k^{tu} K_{[n|qu|} h^s{}_{p]} k^c{}_r \nabla_s h_{tc} \\
&= -k^t{}_u K_{[n}{}^u{}_{|q|} h^s{}_{p]} k^c{}_r \nabla_s h_{tc} \\
&= -K_{[n}{}^u{}_{|q|} h^s{}_{p]} k^t{}_u k^c{}_r \nabla_s h_{tc} \\
&= 0
\end{aligned}$$

Term C:

$$\begin{aligned}
C &= h^m{}_{[n} h^a{}_{p]} h^b{}_q k^c{}_r h^s{}_a h^t{}_b \nabla_m \nabla_s h_{tc} \\
&= h^m{}_{[n} h^s{}_{p]} \left(\nabla_m \nabla_s h_{tc}\right) h^b{}_q h^t{}_b k^c{}_r \\
&= h^m{}_n h^s{}_p \left(\nabla_{[m} \nabla_{s]} h_{tc}\right) h^b{}_q h^t{}_b k^c{}_r \\
&= h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r \left(\nabla_{[m} \nabla_{s]} h_{tc}\right) \\
&= \frac{1}{2} h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r \left(-R^u{}_{tms} h_{uc} - R^u{}_{cms} h_{tu}\right) \\
&= -\frac{1}{2} \left(h^m{}_n h^s{}_p h^b{}_q h^t{}_b \left(k^c{}_r h^u{}_c\right) R_{utms} + h^m{}_n h^s{}_p h^b{}_q h^t{}_b k^c{}_r h^u{}_t R_{ucms}\right) \\
&= -\frac{1}{2} \left(0 + h^m{}_n h^s{}_p h^u{}_q k^c{}_r R_{ucms}\right) \\
&= -\frac{1}{2} h^m{}_n h^s{}_p h^u{}_q k^c{}_r R_{msuc} \\
&= -\frac{1}{2} h^m{}_n h^a{}_p h^b{}_q k^c{}_r R_{mabc}
\end{aligned}$$

In short,

$$h^m{}_{[n}h^a{}_{p]}h^b{}_q k^c{}_r \nabla_m K_{abc} = \frac{1}{2} h^m{}_n h^a{}_p h^b{}_q k^c{}_r R_{mabc}$$

Make another box for this equation.